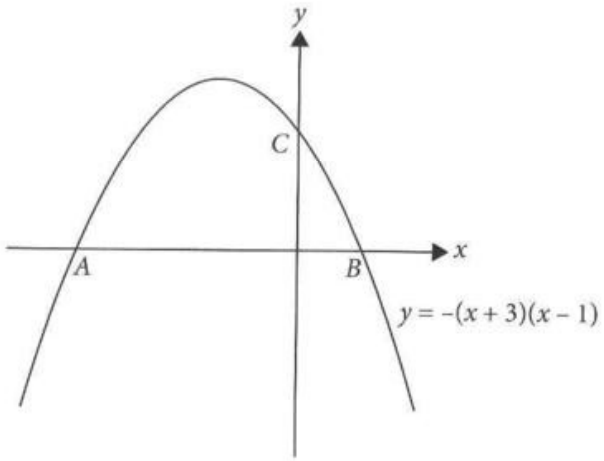
Name: _____ Class: _____ Date: _____

Worksheet 5B

Quadratic functions and graphs

1. The diagram shows the curve $y = -(x + 3)(x - 1)$.
The curve cuts the x-axis at the points A and B, and the y-axis at the point C.

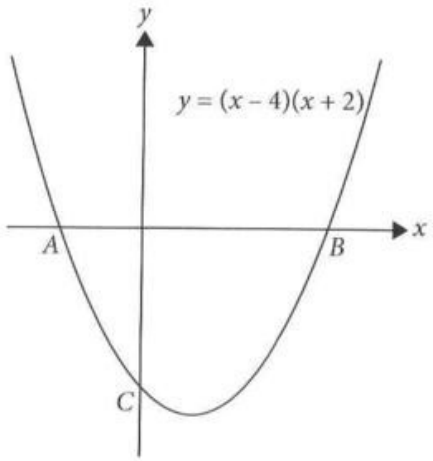


(i) Find the coordinates of A, B and C.

(ii) The equation of the line of symmetry is $x = -1$.
Determine the coordinates of the maximum point of the curve.

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2. In the diagram, the curve $y = (x - 4)(x + 2)$ cuts the x-axis at the points A and B, and the y-axis at the point C.



(i) Find the coordinates of A, B and C.

(ii) The point $P(-3, k)$ lies on the curve.
Find the value of k.

(iii) Explain why the point $Q(3.5, -2.5)$ does not lie on the curve.

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3. The variables x and y are connected by the equation $y = x^2 - 3x - 2$.
Some corresponding values of x and y are given in the table below.

x	−2	−1	0	1	2	3	4
y	p	2	−2	−4	−4	−2	2

(a) Find the value of p.

(b) On the grid, draw the graph of $y = x^2 - 3x - 2$ for $-2 \leqslant x \leqslant 4$.

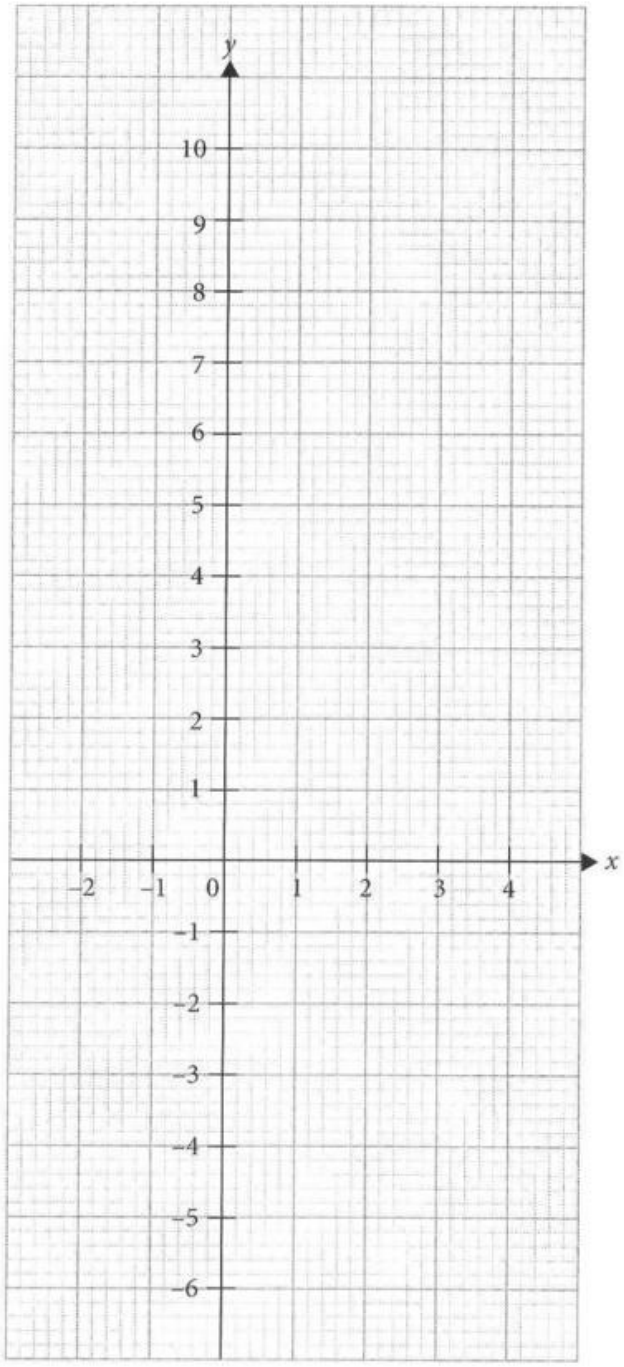
(c) Use your graph in part **(b)** to find

(i) the value of y when $x = 2.8$,

(ii) the values of x when $y = -3$,

(iii) the minimum value of y and the value of x at which this occurs.

(d) State the equation of the line of symmetry of the graph.



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4. The variables x and y are connected by the equation $y = -x^2 + 2x + 1$.
Some corresponding values of x and y are given in the table below.

x	−1	−0.5	0	0.5	1	2	3
y	−2	p	1	1.75	2	1	−2

(a) Find the value of p.

(b) On the grid, draw the graph of $y = x^2 - 3x - 2$ for $-1 \leq x \leq 3$.

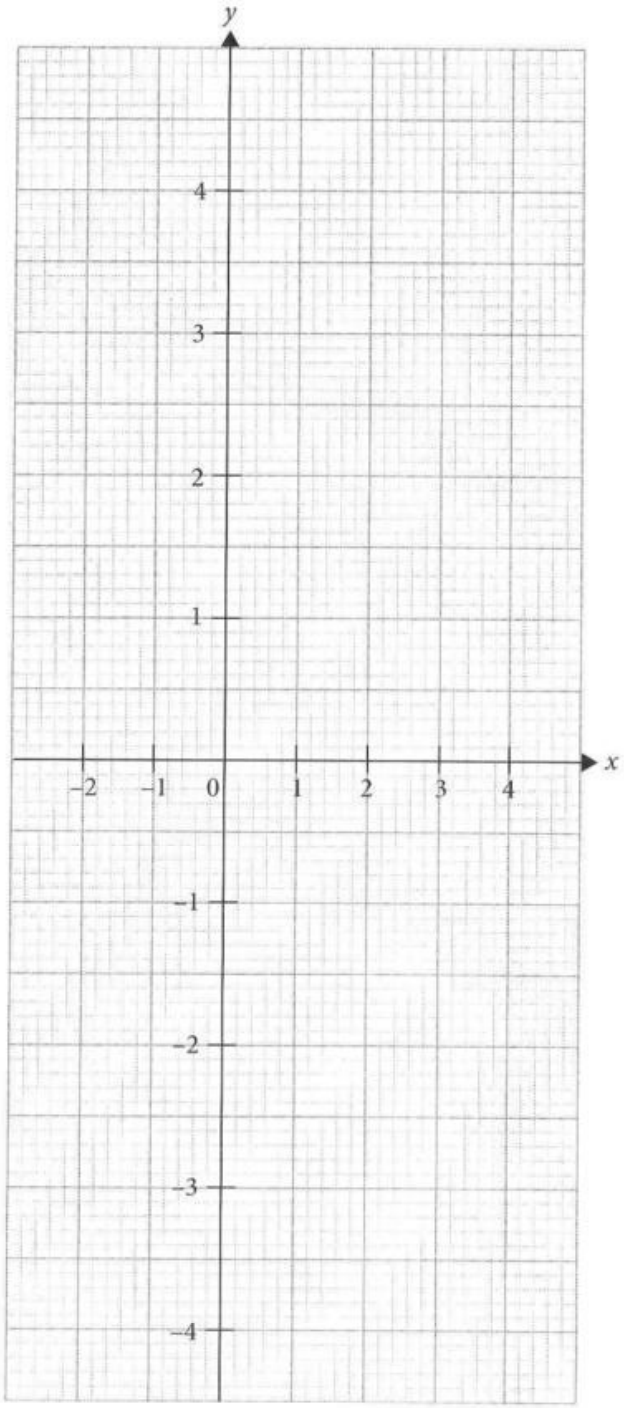
(c) Use your graph in part **(b)** to find

(i) the value of y when $x = -0.7$,

(ii) the values of x when $y = 0.5$,

(iii) the maximum value of y and the value of x at which this occurs.

(d) State the equation of the line of symmetry of the graph.



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5. **(a)** Complete the table.

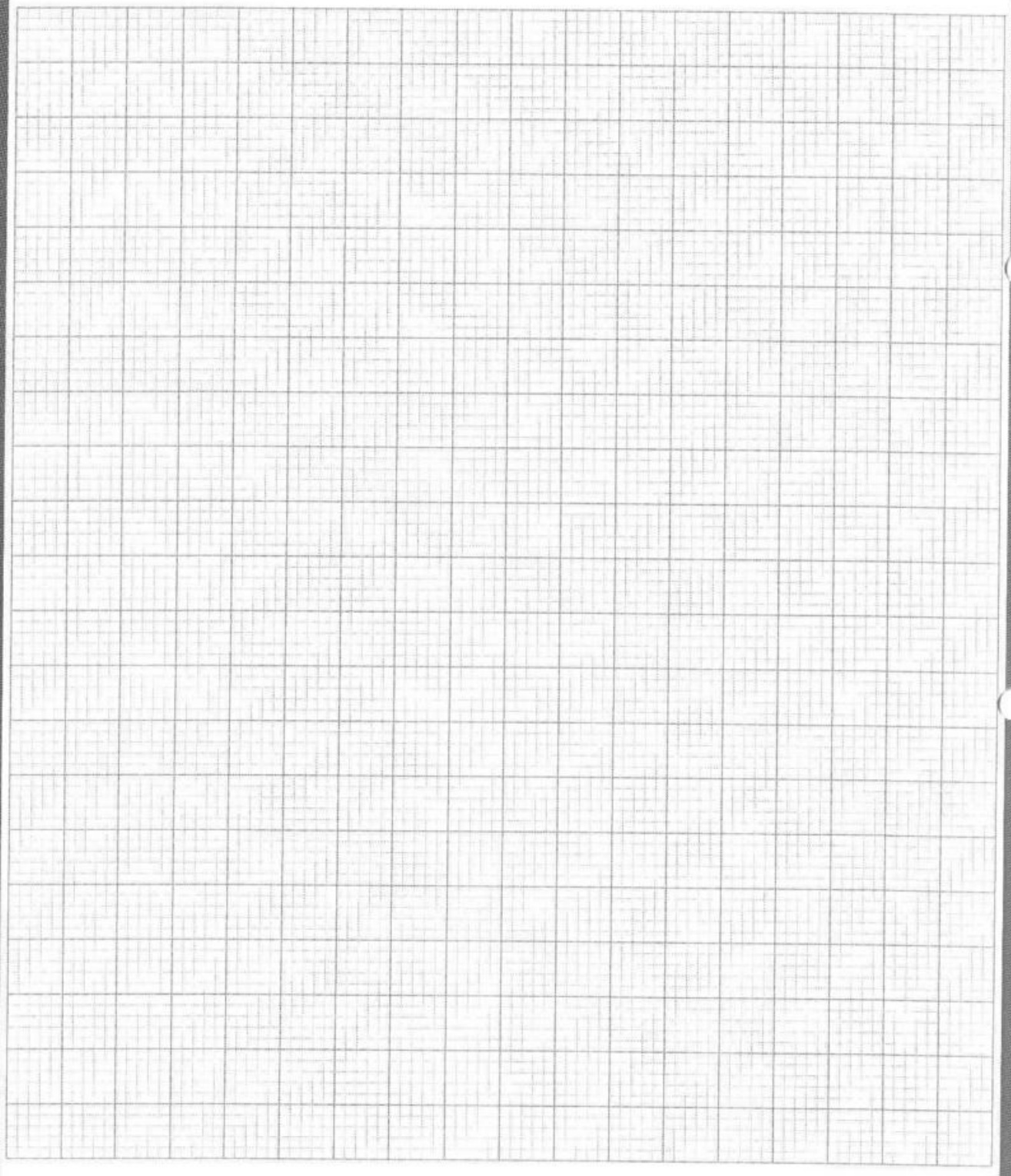
x	−3	−2	−1	0	1	2	3
$y = x^2 - 1$							

(b) Using a scale of 2 cm to represent 1 unit on both axes, draw the graph of $y = x^2 - 1$ for $-3 \leq x \leq 3$.

(c) Use your graph in part **(b)** to find

(i) the coordinates of the minimum point of the graph,

(ii) the equation of the line of symmetry.



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6. **(a)** Complete the table.

x	−2	−1	0	0.5	1	2	3
$y = -2x^2 + x + 6$							

(b) Using a scale of 2 cm to represent 1 unit on the x-axis and 1 cm to represent 1 unit on the y-axis, draw the graph of $y = -2x^2 + x + 6$ for $-2 \leqslant x \leqslant 3$.

(c) Use your graph in part **(a)** to find

(i) the coordinates of the turning point of the graph, stating whether it is a maximum point or a minimum point,

(ii) the equation of the line of symmetry of the graph.

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7. A gardener has 70 metres of fencing to make a rectangular enclosure.
The length of the enclosure is x metres.

(a) Show that the area, A m^2, of the enclosure is given by $A = x(35 - x)$.

(b) This is a table of values for $A = x(35 - x)$.

x	0	5	10	20	30	35
A	0	150	250	p	150	0

(i) Find the value of p.

(ii) Draw the graph of $A = x(35 - x)$.
Use a scale of 2 cm to 5 m on the horizontal x-axis.
Use a scale of 1 cm to 20 m^2 on the vertical A-axis.

(c) Use your graph to find

(i) the value of x when A is a maximum,

(ii) the maximum value of A.

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8. The height, h metres, of a ball projected vertically upwards from a ball machine can be modelled by the equation $h = -4t^2 + 12t$, where t is the time in seconds after it leaves the machine.

(a) **(i)** Solve the equation $-4t^2 + 12t = 0$.

(ii) Write down the duration of flight of the ball.

(b) Draw the graph of $h = -4t^2 + 12t$ during the duration of flight.
Use a scale of 4 cm to 1 s on the horizontal t-axis.
Use a scale of 2 cm to 1 m on the vertical h-axis.

(c) Use your graph to find

(i) the maximum height of the ball above the ground,

(ii) the value(s) of t when the ball is 6 m above the ground.

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9. The shape of an underground passageway can be modelled by the equation $y = -0.02x^2 + 2x - 68$, where x is the horizontal distance in metres from one entrance of the passageway and y is the depth below ground level in metres.

The passageway extends 100 m horizontally.

(a) Draw the graph of $y = -0.02x^2 + 2x - 68$ for $0 \leq x \leq 100$.

Use a scale of 1 cm to 10 m on the horizontal x-axis.

Use a scale of 2 cm to 10 m on the vertical y-axis.

(b) Use your graph to find

(i) whether both entrances of the passageway are located at the same depth,

(ii) the minimum height difference between the passageway and the ground.

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